

Carré du champ Flow Matching

Better Quality-Generalisation Tradeoff in Generative Models

1. Generalisation vs. Memorisation

Generalisation

vs

Memorisation

2. Flow Matching Review

Continuity Equation

$$\frac{\partial}{\partial t} p_t(x) = -\nabla \cdot (u_t(x) p_t(x))$$

Conditional Probability Path

3. The Risk of Memorisation

The $t \rightarrow 1$ Limit

$$p_1(x) \simeq \frac{1}{N} \sum_{i=1}^N \mathcal{N}(x \mid x^{(i)}, \sigma_{min}^2 I)$$

[Figure: Isotropic Blur vs Target Manifold]

4. CDC-FM Conditional Probability Path

Anisotropic & Inhomogeneous Path

$$p_t(x|x_1) = \mathcal{N}(x \mid tx_1, \Sigma_t^\Gamma(x_1))$$

$$\Sigma_t^\Gamma(x_1) = \left[(1-t)I + t\hat{\Gamma}(x_1)^{1/2} \right]^2$$

5. Carré du champ Operator

Local Transition Probabilities

$$P_{ij} = \frac{\exp(-\|x^{(i)} - x^{(j)}\|^2 / \epsilon_i \epsilon_j)}{\sum_k \exp(-\|x^{(i)} - x^{(k)}\|^2 / \epsilon_i \epsilon_k)}$$

Local Covariance ($\hat{\Gamma}$)

$$\hat{\Gamma}(x^{(i)}) = \mathbb{E}_{X \sim P_i} [(X - m^*)(X - m^*)^\top]$$

6. The Algorithm

Target Velocity (Orthogonal Forcing)

$$v_t(x_t|x_1) \leftarrow x_1 - \left(I - \hat{\Gamma}(x_1)^{1/2} \right) x_0$$

Loss Function

7. Experiments

[LiDAR Point Clouds]

[Drosophila Motion]

[Single-Cell Genomics]

[CIFAR-10 Images]

8. Results

[Figure: Quality vs. Generalisation Tradeoff Frontier]

9. Conclusion

$$v_t \perp \mathcal{M}$$

[Figure: Smooth Anisotropic Diffusion]